

# Federated Learning Project

## Personalized Temperature Forecasting for Energy Demand Estimation

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# FL Application

**Task:** predict next-day mean temperature at each FMI weather station  $\Rightarrow$  convert to heating degree days (HDD) as a proxy for regional energy demand.

- ▶ **Predictor:**  $\hat{y}^{(i)} = (\mathbf{x}^{(i)})^\top \mathbf{w}^{(i)}$ ,  
 $\mathbf{x}^{(i)} = [T_{\text{mean}}, T_{\text{min}}, T_{\text{max}}]^\top$
- ▶ **Data:** FMI Open Data, 9 stations, Jan 2022 – Dec 2024

## Why federated:

- ▶ climate varies: coastal (Helsinki) vs. inland (Tampere) vs. sub-arctic (Inari)
- ▶ single global model: too coarse
- ▶ purely local model: unreliable when data is scarce
- ▶ neighbours share weather regime  $\Rightarrow$  collaborate

# FL Network

**Nodes:**  $n = 9$  FMI stations (coastal / inland / northern Finland)

**Local data per station  $i$ :**

$$\mathcal{D}^{(i)} = \{(\mathbf{x}^{(i,t)}, y^{(i,t)})\}_{t=1}^{m_i}, \quad y^{(i,t)} = T_{\text{mean}}^{(t+1)}$$

**Two FL systems - same nodes, different edges:**

► **System A - Distance graph:**

$$A_{i,i'} = \exp\left(-\frac{d^2}{2\sigma^2}\right), \quad d \leq 300 \text{ km}, \quad \sigma = 150 \text{ km}$$

► **System B - Correlation graph:**

$$A_{i,i'}^{(B)} = \max\left(0, \rho(T^{(i)}, T^{(i')})\right) \cdot \exp\left(-\frac{d^2}{2\sigma^2}\right)$$

Both systems: 15 edges, weights 0.15–0.82. All pairs within 300 km have positive correlation.

# GTVMin: Objective

**Variation measure** (squared- $\ell_2$ ):

$$d(\mathbf{w}^{(i)}, \mathbf{w}^{(i')}) = \|\mathbf{w}^{(i)} - \mathbf{w}^{(i')}\|_2^2$$

**Full GTVMin objective:**

$$\min_{\mathbf{w}^{(1)}, \dots, \mathbf{w}^{(n)}} \underbrace{\sum_{i=1}^n L_i(\mathbf{w}^{(i)})}_{\text{local fit}} + \alpha \sum_{\{i, i'\} \in \mathcal{E}} A_{i, i'} \|\mathbf{w}^{(i)} - \mathbf{w}^{(i')}\|_2^2 \text{GTV penalty}$$

- ▶  $\alpha = 0$ : independent local training (Baseline)
- ▶  $\alpha \rightarrow \infty$ : single global model
- ▶  $\alpha^*$ : tuned on validation set per system

# FL Algorithm: Ridge Reformulation

**Key insight:** GTVMin subproblem for node  $i$  (all others fixed) is a Ridge regression.

**Neighbour aggregate:**

$$s_i = \sum_{i' \in \mathcal{N}(i)} A_{i,i'}, \quad \bar{\theta}_i = \frac{1}{s_i} \sum_{i' \in \mathcal{N}(i)} A_{i,i'} \mathbf{w}^{(i')}$$

**Substitution**  $\mathbf{u} = \mathbf{w}^{(i)} - \bar{\theta}_i \Rightarrow$  plain Ridge:

$$\min_{\mathbf{u}} \left\| \mathbf{X}^{(i)} \mathbf{u} - \tilde{\mathbf{y}}^{(i)} \right\|_2^2 + \alpha s_i \left\| \mathbf{u} \right\|_2^2, \quad \tilde{\mathbf{y}}^{(i)} = \mathbf{y}^{(i)} - \mathbf{X}^{(i)} \bar{\theta}_i$$

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Ridge(alpha= $\alpha s_i$ , fit_intercept=False)
.fit(X, y_tilde)  $\Rightarrow \mathbf{w}^{(i)} = \mathbf{u}^* + \bar{\theta}_i$ 
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**Synchronous**,  $T = 50$  rounds,  $\mathbf{w}^{(i,0)} = \mathbf{0}$ . No learning rate needed - closed-form solution.

# Experiments: Setup & Key Finding

**Three experiments** - same systems, less training data:

| Exp.        | Train                               | Val     | Test    |
|-------------|-------------------------------------|---------|---------|
| 1 - Full    | 2022–2023 ( $\approx 730\text{d}$ ) | 2024 H1 | 2024 H2 |
| 2 - Reduced | 2022 ( $\approx 365\text{d}$ )      | 2023 H1 | 2023 H2 |
| 3 - Minimal | 2022 H1 ( $\approx 181\text{d}$ )   | 2022 Q3 | 2022 Q4 |

$\alpha$  **grid:**  $\{10^{-4}, 10^{-3}, \dots, 500\}$ , tuned on validation set per system.

**Baseline:** local-only Ridge ( $\alpha = 0$ , no FL).

**Splits:** chronological - no shuffling (time series constraint).

# Results

Mean test RMSE (°C) across 9 stations:

| System   | Exp. 1 | Exp. 2 | Exp. 3      |
|----------|--------|--------|-------------|
| Baseline | 5.29   | 5.47   | 4.15        |
| System A | 5.30   | 5.47   | <b>4.07</b> |
| System B | 5.30   | 5.47   | <b>4.07</b> |

- ▶ **Exp. 1 & 2:** FL makes no difference - local data sufficient for 3-feature linear model
- ▶ **Exp. 3:** FL helps - 2.1% RMSE reduction with only 181 training days
- ▶ Largest gains: Tampere, Kuopio, Jyväskylä (inland)
- ▶ System A  $\approx$  System B - distance is sufficient proxy for climate similarity

# Conclusion

## Key finding:

- ▶ FL helps only when local data is scarce - 2.1% RMSE reduction with 181 training days, no benefit with 365+ days
- ▶ Distance graph  $\approx$  correlation graph - all station pairs positively correlated
- ▶ HDD over-estimated at southern stations in summer

## Future work:

- ▶ Lagged/polynomial features  $\Rightarrow$  more FL benefit
- ▶ Seasonal models  $\Rightarrow$  fix summer HDD bias

**Thank you - questions?**